



CONTACT

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AT A GLANCE

20 years' experience	3 continents
10 streaming & AI projects	20 data & BI projects

CORE SKILLS

Cloud: AWS (Kinesis, S3, Redshift, Lambda, ECS)
· Google Cloud (BigQuery, Looker, Airflow, Dataflow)

Data & modelling: SQL · DBT · dimensional / semantic-layer modelling · data warehouse & data/Delta Lake · GraphQL data services

Integration / orchestration: Apache Airflow · Apache Hop · CDC · Greengrass · Terraform Docker · CI/CD · Python

Analytics & visualization: Tableau · Looker · Power BI · Elasticsearch/Kibana

Applied AI: RAG ingestion & embedding · agentic AI · LLM for unstructured messages

Consulting: Architecture sizing & pricing · technical pre-sales · training & enablement · problem determination

TRAINING — 200+ trained

Big Data, streaming & IoT architectures: Reply, Nordcom, Leonardo, Spindox, Everis, BTicino, Zain, DEWA, Vodafone Egypt
Cloud architectures & migration: IngramMicro, Unicredit
Data visualization (Tableau, Looker, OSS): TIM, Leonardo, Injenia
Big Data / Cloudera data migration: Banque du Mozambique

EDUCATION

Bachelor of Engineering, Computer Engineering — University of Bologna, 1996–2003 — 100/100 with merit.

LANGUAGES

English (professional) · Italian (native)

BEYOND WORK

I run creative robotics and technology labs for kids in schools: hands-on, playful learning. It's the same instinct that shapes my consulting — making technology understandable and helping people actually use it.

Virgilio Pierini

Senior Data & Cloud Architect & Trainer

-Analytics engineering & making data AI-ready-

Senior data and cloud consultant with 20+ years designing and building data platforms end to end — data integration, dimensional and semantic modelling, data warehouses and lakes on GCP and AWS, and analytics with Tableau, Looker and Power BI. My focus is the modelled layer and data services beneath the dashboards: the foundation that makes both BI and today's conversational, AI-driven analytics reliable. Recently hands-on in applied AI on real business data (RAG pipelines, tool-using agents). Strong in architecture sizing, technical pre-sales and training. Fluent English, international experience.

Objective

My role is to help organizations design, size and price the right data architecture — and get their teams to adopt it — bridging BI, semantic-layer modelling and applied AI (RAG, agents). Remote, with on-site delivery for design and enablement.

Selected projects

Data Integration & AI Help-desk Agent Ympact — Fashion & Luxury — Italy — 2026

Data integration with Apache Hop (orders, shipments); designed and built a tool-using AI help-desk agent that traces orders through application logs — moving business operators from dashboards to asking questions in natural language.

Event-driven trading with hybrid LLM parsing Personal project — 2025

Event-driven pipeline with a deterministic, guard-railed state machine mediating between LLM-parsed human input and an order-execution system. Hybrid regex+LLM parser benchmarked at ~98% accuracy on hundreds of real human-written messages; run on a demo account to empirically measure signal reliability. Cost < \$10/month.

Time-series Data Science for Transportation A10 — Motorway — Italy — 2023

What-if platform estimating how human-controlled factors (maintenance, diversions) impact travel time, to plan works and reduce traffic impact. Time-series data science with Python notebooks and Tableau reporting, alongside a senior data scientist.

Data Solutions on Google Cloud Injenia — Google Partner — Italy — 2020–Dec 2025

Design of data solutions, pipelines and integrations for end customers migrating to GCP (BigQuery, Airflow, Beam, CDAP), plus data-modelling and visualization enablement (Looker, Tableau).

Big Data Architecture & Modelling for Social Media TaTaTu — Social entertainment platform — Italy/Serbia — 2022

Coordinated a small team on Terraform infrastructure, Beam pipelines and Python CDC extractors. Modelled event ingestion and the feature table for the recommendation engine; ML models in production (BigQuery ML + AWS). Kinesis → S3 → Redshift, ELK.

Streaming Analytics, Industry 4.0 / e-mobility Reinova — Battery testing — Italy — 2021–2022

Modern Data Stack on AWS (Delta Lake, dbt, Airflow) with edge collection via AWS Greengrass. Architecture sized from scratch to several TB; optimization of electrical-test scheduling.

Streaming Analytics & Data Services, Industry 4.0 AVL — Engine manufacturing — Italy — 2018–2020

Log integration and text extraction into type-2 fact tables and entity-centric data in Elasticsearch. Designed GraphQL data services beneath custom dashboards (D3 / Chart.js). Full ELK stack on AWS.

Telco Data Lake on Cloudera Zain — Telecommunications — Kuwait — 2017–2018

Data lake from siloed Oracle systems to a Cloudera Hadoop stack: 20+ source systems, 1 PB/year. Ingestion and processing with Spark, enriching an existing SAP HANA solution.

Big Data Platform for Media RedPepper / Big Brother — South Africa — 2016

Designed a 5 PB big-data Lambda architecture for transcript analysis to select the most relevant advertisement.

Professional experience

LLM Data Quality / RAG — Chief Data Officer You-AI — UAE — 2024–Aug 2025

RAG tuning, LLM orchestration and retrieval for a consumer application.

Big Data & Visualization — Senior Consultant Injenia — Google Partner — Italy — 2020–2025

BI and data solutions for Big Data projects; Google Cloud architectures; trainer for data modelling and visualization.

Big Data & Cloud Solutions / Evangelist Clickode MEA (IBM & Microsoft partner) — MEA — 2015–2017

Big Data & analytics for the private sector; MAAS/IAAS/SAAS/PAAS on IBM Bluemix, Microsoft Azure/Power BI, SAP HANA; technical pre-sales, solution design and deployment.

Data Architect Cineca — Higher Education Consortium — Italy — 2007–2016

Team leader for the new Data Scientists office; Hortonworks/BigInsights; Spark/R; data warehouse & system administration; Hyperion and Pentaho architect.

Datawarehouse Specialist CSE — Centro Servizi Bancari — Italy — 2003–2007

Enterprise data-warehouse projects; analysis, modelling and development in Hyperion; dev team leadership; hosting-service infrastructure.

Earlier projects (selection)

CRM cloud data integration — CNH (Industrial, Europe, 2018–2019, Azure CRM/SQL Server/SSIS) · SAP Leonardo IoT for waste management (Utilities, Italy, 2018) · Cloud migration — Fondazione Cariplo (Finance, 2017–2018, Azure) · Fleet management on OBDII + Power BI — FleetMonitor24 (2016–2018) · Spark modelling layer over legacy DW — Comune di Torino (Public, 2017) · AWS Lambda + DynamoDB for data science — Prometeia (Finance, 2018) · Sentiment analysis, Mille Miglia — Comune di Brescia (2016) · IoT predictive maintenance — Alstom (Railway, 2016) · Time-series data-quality validation — Arpa Emilia-Romagna (Climate, 2015–2016).